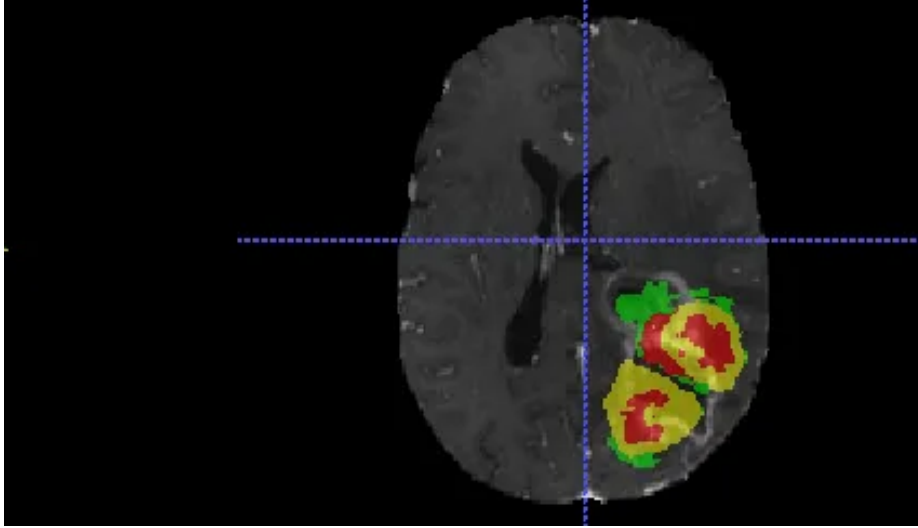




Original MRI Image credit: TCIA

Medical image segmentation refers to the demarcation of a patient's scan (such as a CT or MRI scan) to show the extent of the disease, so accurate diagnosis and treatment can be made. Expert radiologists have typically done this manually, which is time-intensive and potentially subject to observer bias. The use of deep learning networks for medical image segmentation has come a long way in the last decade, with significant increases in sophistication and accuracy.

When I started on an internship project to segment Meningeal tumors using DNNs, I was very new to this area and went through a steep learning process. Here, I am sharing my journey in a series of posts that walk you through the entire project on a similar, publically accessible, step by step.

In this series, we will train a DNN using the nnU-Net framework for automated segmentation of de-identified Glioblastoma brain MRI scans. Glioblastoma is an aggressive, malignant brain tumor with a (<6%) survival rate. Using deep learning to accurately segment

hope you follow along.

All the code for this project is available on my [Github page](#)

begin writing the code. To keep costs down, we will use Colab until we begin the actual model training.

Part 4 — Training and Test Data Setup: Here, we split the data into training and test cohorts (80/20 proportion) and use `os` to move the MRI scans to the appropriate folders for preprocessing and inference.

Part 5 — Data Preprocessing: Before training the model, we preprocess the MRI scans to normalize shape, intensity and other characteristics. We do that here using the `nnUNet` package. We also write some code to process the ground truth segmentation labels for training.

Part 6 — Neural Network Model Training: Finally, all the configuration is complete, and we can begin training the model. This will require more compute resources than what's available in the free version. I signed up for Google's Colab Pro "Pay as you go" plan for 100 compute units for \$9.99, which was sufficient for this part.

Part 7 — Inference: Once the DNN is trained, we use the model to make predictions on demarcating the tumor region segmentation in the MRI scans in the test cohort. We also learn how to evaluate model performance.

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Written by Diya S

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I am a high school student on a journey to understand how disruptive technologies can transform and democratize healthcare in the coming D.S

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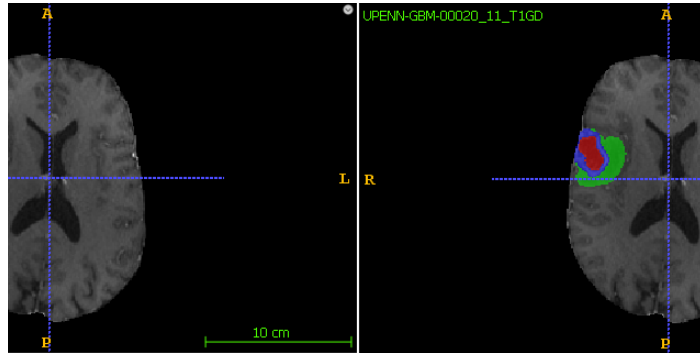
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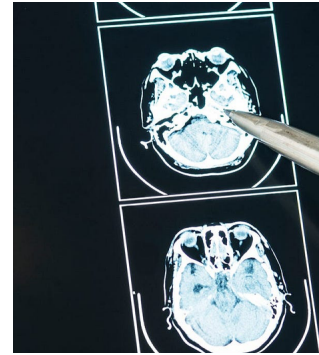


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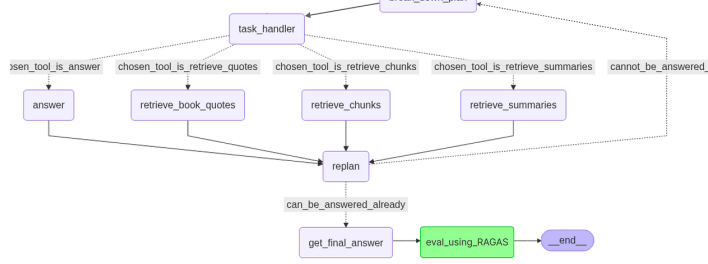
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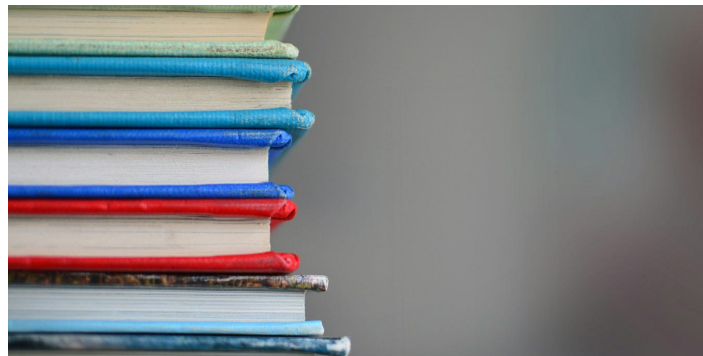
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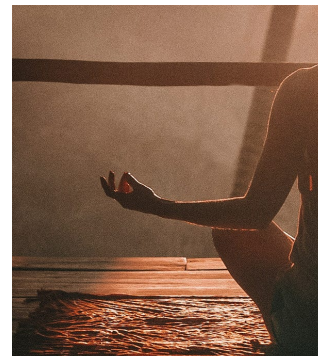


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